Logical Reasoning – Lesson 1

Section A: Syllogism

Arguments are a set of statements(**premises & conclusion**). In other words, conclusions and premises are articulated in the form of statements. Statements are sentences that can be determined to possess or lack truth.

A **conclusion/ inference** is a statement that is being argued for.

A **premise(proposition)** is a statement that substantiates a conclusion. It provides evidence, reasons and grounds for the conclusion and therefore supports/ justifies the conclusion. There can be more than one premise.

The phenomenon of deriving a conclusion from a given premise or a set of premises is known as **logical deduction.**An argument attempts to draw some logical connection between the premise(s) and the conclusion.Therefore, logical deduction can be viewed as a process of reasoning from the truth of the premises to the truth of the conclusion.

Please note that for a conclusion to be valid, it must be definitely true based on the given premise(s). A statement which may or may not be true is not a valid conclusion.

A **syllogism** is a three-part logical argument based on deductive reasoning, in which two or more premises are combined to arrive at a conclusion.

Example: -

Premises:
All men are dogs.
All dogs are cats.

Conclusion: All men are cats.

Exercise: -

In questions 1-5 below, two statements followed by two conclusions numbered I and II have been given. You have to consider the given statements as true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follow from the given statements, disregarding commonly known facts.

1. Statements: All drums are flutes.
   All guitars are drums.

   Conclusions: All guitars are flutes.
   All drums are guitars.

   a. Both conclusions I and II are true
   b. Only conclusion I is true
   c. Neither conclusion I nor II is true
   d. Only conclusion II is true
   e. Either conclusion I or II is true

2. Statements: Some speakers are guests.
   No guest is a dignitary.

Conclusions: No dignitary is a speaker.
Some guests are definitely not speakers.

a. Only conclusion II is true
b. Both conclusions I and II are true
c. Either conclusion I or II is true
d. Neither conclusion I nor II is true
e. Only conclusion I is true

3. Statements: All spades are tools.
No spade is a vessel.

Conclusions: At least some tools are vessels.
No tool is a vessel.

a. Either conclusion I or II is true
b. Neither conclusion I nor II is true
c. Only conclusion I is true
d. Both conclusions I and II are true
e. Only conclusion II is true

4. Statements: All users are senders.
Some users are machines.

Conclusions: Some senders are machines.
All senders are machines.

a. Only conclusion I is true
b. Only conclusion II is true
c. Either conclusion I or II is true
d. Neither conclusion I nor II is true
e. Both conclusions I and II are true

5. Statements: Some cooks are employees.
Some employees are workers.

Conclusions: All cooks are workers.
No cook is a worker.

a. Either conclusion I or II is true
b. Neither conclusion I nor II is true
c. Only conclusion I is true
d. Both conclusions I and II are true

e. Only conclusion II is true

In questions 6-10 below, two statements followed by two conclusions numbered I and II have been given. You have to consider the given statements as true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follow from the given statements, disregarding commonly known facts.

Give answer
a. If only conclusion I follows
b. If only conclusion II follows
c. If either conclusion I or II follows
d. If neither conclusion I nor II follows

e. If both conclusions I and II follow

6. Statements: All water is divine.
All temples are divine.

Conclusions: I. All water is temple.
II. All temples are water.

7. Statements: All jungles are tigers.
Some tigers are horses .

Conclusions: I. Some horses are jungles.
II. No horse is jungle.

8. Statements: All roads are waters.
Some waters are boats.

Conclusions: I. Some boats are roads.
II. All waters are boats.

9. Statements: Some papers are pens.
Some pencils are pens.

Conclusions: I. Some pens are pencils.
II. Some pens are papers.

10. Statements: All windows are doors.
No door is wall.

Conclusions: I. No window is wall.
II. No wall is door.

Section B: Statement &Conclusions

Sometimes we may be required to derive/ deduce conclusions from the contents of a given sentence/ statement or a given paragraph. In such cases we must analyse the given statement(s), understand their direct/ indirect implications and then decide which of the conclusions follows logically and definitely/ certainly, from these statement(s).

Example: -

Statement: This book 'Z' is the only book which focuses its attention to the problem of poverty in India between 1950 and 1980.

Conclusion: No other book deals with poverty in India during 1950 to 1980.

Exercise: -

In questions 1-10 below is given a statement followed by two conclusions numbered I and II. You have to assume everything in the statement to be true, then consider the two conclusions together and decide which of them logically follows beyond a reasonable doubt from the information given in the statement.
Give answer

a. If only conclusion I follows
b. If only conclusion II follows
c. If either conclusion I or II follows
d. If neither conclusion I nor II follows

e. If both conclusions I and II follow

1. The percentage of the national income shared by the top 10 per cent of households in India is 35.

   I. When an economy grows fast, concentration of wealth in certain pockets of the population takes place.
   II. The national income is unevenly distributed in India.

2. The Prime Minister emphatically stated that his government will make every possible effort for the upliftment of poor farmers and farmhands.

   I. Except poor farmers and farmhands, all others have got benefits of the fruits of development.
   II. No serious efforts have been made in the past for upliftment of any section of the society.

3. The Cabinet of State X took certain steps to tackle the milk glut in the state as the cooperatives and government dairies failed to use the available milk – A news report.

   I. The milk production of State X is more than its need.
   II. The government and cooperative dairies in State X are not equipped in terms of resources and technology to handle such excess milk.

4. Until our country achieves economic equality, political freedom and democracy would be meaningless.

   I. Political freedom and democracy go hand in hand.
   II. Economic equality leads to real political freedom and democracy.

5. National Aluminium Company has moved India from a position of shortage to self-sufficiency in the metal.

   I. Previously, India had to import aluminium.
   II. With this speed, it can soon become a foreign exchange earner.

6. Prime age school-going children in urban India have now become avid as well as more regular viewers of television, even in households without a TV. As a result, there has been an alarming decline in the extent of readership of newspapers.

   I. Method of increasing the readership of newspapers should be devised.
   II. A team of experts should be sent to other countries to study the impact of TV on the readership of newspapers.

7. In a recent survey report, it has been stated that those who undertake physical exercise for at least half-an hour a day are less prone to have any heart ailments.

   I. Moderate level of physical exercise is necessary for leading a healthy life.
   II. All people who do desk-bound jobs definitely suffer from heart ailments.

8. This world is neither good nor evil; each person manufactures a world for herself/ himself.

   I. Some people find this world quite good.
   II. Some people find this world quite bad.

9. Recent trends also indicate that the number of child migrants in large cities is increasing. These children leave their families to join the ranks of urban poor doing odd jobs in markets, workshops, hotels or in service sectors.

I. Migration to big cities should be checked.
II. The plight of poor children should be thoroughly studied.

10. Today, out of the world population of several thousand million, the majority of men have to live under governments which refuse them personal liberty and the right to dissent.

    I. People are indifferent to personal liberty and the right to dissent.
    II. People desire personal liberty and the right to dissent.

Section C: Statement & Assumptions

An assumption is the underlying information that must hold true for some given statement(s) to be valid. It can be inferred by analysing the given statement(s). Assumptions are never stated explicitly but are implied by the given statement(s).

Please note that an assumption need not be definitely true. It could very well state the possibility of some anticipated event.

Example: -

Vitamin E tablets improve circulation, keeping your complexion in a glowing condition.

The two underlying assumptions are: -

People like a glowing complexion.
Complexion becomes dull in the absence of circulation.

Exercise: -

In questions 1-10 below is given a statement followed by two assumptions numbered I and II. You have to consider the statement and the following assumptions and decide which of the assumptions is implicit in the given statement.

Give answer
a. If only assumption I is implicit
b. If only assumption II is implicit
c. If either assumption I or II is implicit
d. If neither assumption I nor II is implicit
e. If both assumptions I and II are implicit

1. "I have not received telephone bills for nine months inspite of several complaints." – A telephone customer's letter to the editor of a daily.

    I. Every customer has a right to get bills regularly from the telephone company.
    II. The customer's complaints point to the defect in the services which is expected to be corrected.

2. Most people who stop smoking gain weight.

    I. If one stops smoking, one will gain weight.
    II. If one does not stop smoking, one will not gain weight.

3. Nobody can predict as to how long our country would take to contain the unfortunate and disastrous terrorist activities.

I. It is impossible to put an end to terrorist activities.
II. Efforts to control the terrorist activities are on.

4. "Private property, trespassers will be prosecuted." – A notice on a plot of land.

I. The passer-by may read the notice and may not trespass.
II. The people are scared of prosecution.

5. The organisation should promote employees on the basis of merit alone and not on the basis of length of service or seniority.

I. Length of service or seniority does not alone reflect merit of an employee.
II. It is possible to measure and determine merit of an employee.

6. The present examination system needs overhauling thoroughly.

I. The present examination system is obsolete.
II. Overhauling results in improvement.

7. Kartik left for Delhi on Tuesday by train to attend a function to be held on Friday at his uncle's house in Delhi.

I. Kartik may reach Delhi on Wednesday.
II. Kartik will reach Delhi by Friday.

8. "Ensure a good night's sleep for your family for your family with safe and effective X mosquito coil." – An advertisement.

I. X mosquito coil is better than any other mosquito coil.
II. A good night's sleep is desirable.

9. The Parent Teacher Association(PTA) of a school has informed the Principal that they will not send their children to the school unless the school authority reduces the fees with immediate effect.

I. Majority of the parents may agree with the PTA and may not send their wards to the school.
II. The school authority may accede to the demand of the PTA and reduce the fees.

10. The government has set up a fact finding mission to look into the possible reasons for the recent violence in the area.

I. The nation may be able to come up with credible information about the incidents.
II. The people in the area may cooperate with the mission and come forward to give detailed information related to the incidents.